



AP[®] Calculus BC Practice Exam

From the 2012 Administration

Revised January 2013

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CALCULUS BC
SECTION I, Part A
Time—55 minutes
Number of questions—28

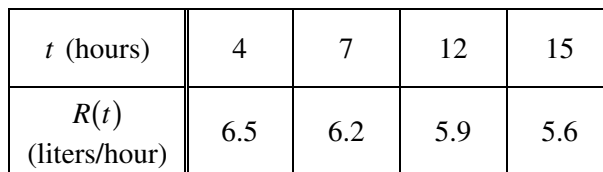
Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

1. If $y = \sin^3 x$, then $\frac{dy}{dx} =$

2. The position of a particle moving in the xy -plane is given by the parametric equations $x(t) = t^3 - 3t^2$ and $y(t) = 12t - 3t^2$. At which of the following points (x, y) is the particle at rest?

- 4-



- $$\text{I. } \sum_{n=1}^{\infty} \frac{8^n}{n!}$$

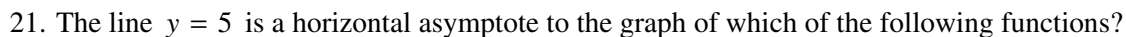
$$\text{II. } \sum_{n=1}^{\infty} \frac{n!}{n^{100}}$$

$$\text{III. } \sum_{n=1}^{\infty} \frac{n+1}{(n)(n+2)(n+3)}$$

- 8-

18. The graph of f' , the derivative of a function f , consists of two line segments and a semicircle, as shown in the figure above. If $f(2) = 1$, then $f(-5) =$

- (A) $2\pi - 2$
(B) $2\pi - 3$
(C) $2\pi - 5$
(D) $6 - 2\pi$
(E) $4 - 2\pi$



22. The power series $\sum_{n=0}^{\infty} a_n(x-3)^n$ converges at $x = 5$. Which of the following must be true?

- 16-**

25. $\int_1^{\infty} xe^{-x^2} dx$ is

- 18-

27. For what values of p will both series $\sum_{n=1}^{\infty} \frac{1}{n^{2p}}$ and $\sum_{n=1}^{\infty} \left(\frac{p}{2}\right)^n$ converge?

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any part of this page is illegal.

28. Let g be a continuously differentiable function with $g(1) = 6$ and $g'(1) = 3$. What is $\lim_{x \rightarrow 1} \frac{\int_1^x g(t) dt}{g(x) - 6}$?

B**B****B****B****B****B****B****B****B**

CALCULUS BC
SECTION I, Part B
Time—50 minutes
Number of questions—17

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON
THIS PART OF THE EXAM.

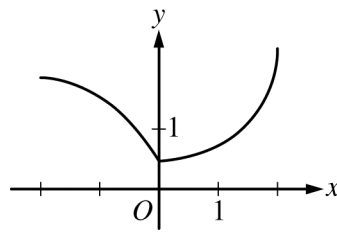
Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–92.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

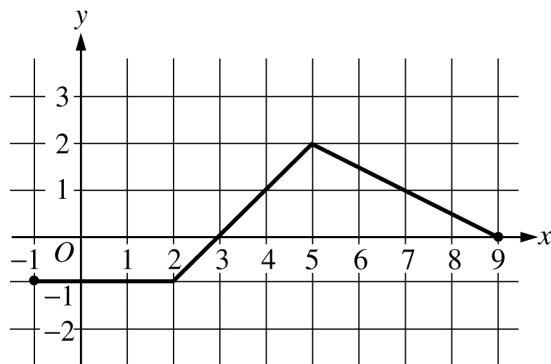
- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

B**B****B****B****B****B****B****B****B**

Graph of f

76. The function f , whose graph is shown above, is defined on the interval $-2 \leq x \leq 2$. Which of the following statements about f is false?
- (A) f is continuous at $x = 0$.
 - (B) f is differentiable at $x = 0$.
 - (C) f has a critical point at $x = 0$.
 - (D) f has an absolute minimum at $x = 0$.
 - (E) The concavity of the graph of f changes at $x = 0$.

-
77. Let f and g be the functions given by $f(x) = e^x$ and $g(x) = x^4$. On what intervals is the rate of change of $f(x)$ greater than the rate of change of $g(x)$?
- (A) $(0.831, 7.384)$ only
 - (B) $(-\infty, 0.831)$ and $(7.384, \infty)$
 - (C) $(-\infty, -0.816)$ and $(1.430, 8.613)$
 - (D) $(-0.816, 1.430)$ and $(8.613, \infty)$
 - (E) $(-\infty, \infty)$

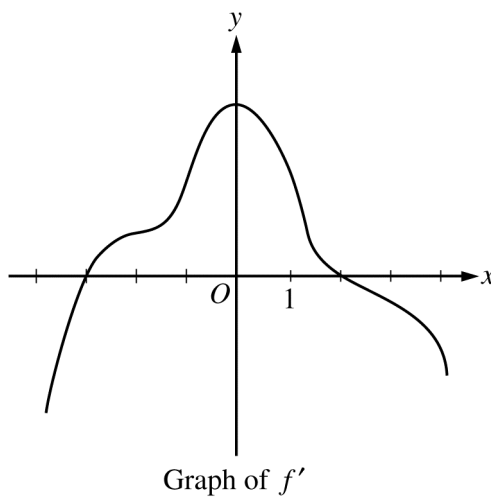
B**B****B****B****B****B****B****B****B**Graph of f

78. The graph of the piecewise linear function f is shown above. What is the value of $\int_{-1}^9 (3f(x) + 2) dx$?

- (A) 7.5 (B) 9.5 (C) 27.5 (D) 47 (E) 48.5

79. Let f be a function having derivatives of all orders for $x > 0$ such that $f(3) = 2$, $f'(3) = -1$, $f''(3) = 6$, and $f'''(3) = 12$. Which of the following is the third-degree Taylor polynomial for f about $x = 3$?

- (A) $2 - x + 6x^2 + 12x^3$
(B) $2 - x + 3x^2 + 2x^3$
(C) $2 - (x - 3) + 6(x - 3)^2 + 12(x - 3)^3$
(D) $2 - (x - 3) + 3(x - 3)^2 + 4(x - 3)^3$
(E) $2 - (x - 3) + 3(x - 3)^2 + 2(x - 3)^3$

B**B****B****B****B****B****B****B****B**

Graph of f'

80. The graph of f' , the derivative of the function f , is shown above. Which of the following statements must be true?

I. f has a relative minimum at $x = -3$.

II. The graph of f has a point of inflection at $x = -2$.

III. The graph of f is concave down for $0 < x < 4$.

(A) I only (B) II only (C) III only (D) I and II only (E) I and III only

B**B****B****B****B****B****B****B****B**

	$0 < x < 1$	$1 < x < 2$
$f(x)$	Positive	Negative
$f'(x)$	Negative	Negative
$f''(x)$	Negative	Positive

81. Let f be a function that is twice differentiable on $-2 < x < 2$ and satisfies the conditions in the table above. If $f(x) = f(-x)$, what are the x -coordinates of the points of inflection of the graph of f on $-2 < x < 2$?
- (A) $x = 0$ only
(B) $x = 1$ only
(C) $x = 0$ and $x = 1$
(D) $x = -1$ and $x = 1$
(E) There are no points of inflection on $-2 < x < 2$.
-

82. What is the average value of $y = \sqrt{\cos x}$ on the interval $0 \leq x \leq \frac{\pi}{2}$?

(A) -0.637 (B) 0.500 (C) 0.763 (D) 1.198 (E) 1.882

B**B****B****B****B****B****B****B****B**

83. If the function f is continuous at $x = 3$, which of the following must be true?

- (A) $f(3) < \lim_{x \rightarrow 3} f(x)$
 - (B) $\lim_{x \rightarrow 3^-} f(x) \neq \lim_{x \rightarrow 3^+} f(x)$
 - (C) $f(3) = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$
 - (D) The derivative of f at $x = 3$ exists.
 - (E) The derivative of f is positive for $x < 3$ and negative for $x > 3$.
-

84. For $-1.5 < x < 1.5$, let f be a function with first derivative given by $f'(x) = e^{(x^4 - 2x^2 + 1)} - 2$. Which of the following are all intervals on which the graph of f is concave down?

- (A) $(-0.418, 0.418)$ only
- (B) $(-1, 1)$
- (C) $(-1.354, -0.409)$ and $(0.409, 1.354)$
- (D) $(-1.5, -1)$ and $(0, 1)$
- (E) $(-1.5, -1.354)$, $(-0.409, 0)$, and $(1.354, 1.5)$

B**B****B****B****B****B****B****B****B**

85. The fuel consumption of a car, in miles per gallon (mpg), is modeled by $F(s) = 6e^{\left(\frac{s}{20} - \frac{s^2}{2400}\right)}$, where s is the speed of the car, in miles per hour. If the car is traveling at 50 miles per hour and its speed is changing at the rate of 20 miles/hour², what is the rate at which its fuel consumption is changing?

- (A) 0.215 mpg per hour
- (B) 4.299 mpg per hour
- (C) 19.793 mpg per hour
- (D) 25.793 mpg per hour
- (E) 515.855 mpg per hour

B**B****B****B****B****B****B****B****B**

86. If $f'(x) > 0$ for all real numbers x and $\int_4^7 f(t) dt = 0$, which of the following could be a table of values for the function f ?

(A)

x	$f(x)$
4	-4
5	-3
7	0

(B)

x	$f(x)$
4	-4
5	-2
7	5

(C)

x	$f(x)$
4	-4
5	6
7	3

(D)

x	$f(x)$
4	0
5	0
7	0

(E)

x	$f(x)$
4	0
5	4
7	6

B**B****B****B****B****B****B****B****B**

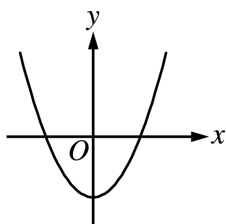
87. Let R be the region in the first quadrant bounded above by the graph of $y = \ln(3 - x)$, for $0 \leq x \leq 2$. R is the base of a solid for which each cross section perpendicular to the x -axis is a square. What is the volume of the solid?

(A) 0.442 (B) 1.029 (C) 1.296 (D) 3.233 (E) 4.071

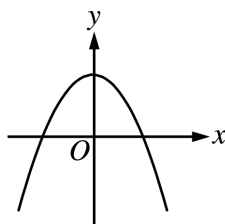
B**B****B****B****B****B****B****B****B**

88. The derivative of a function f is increasing for $x < 0$ and decreasing for $x > 0$. Which of the following could be the graph of f ?

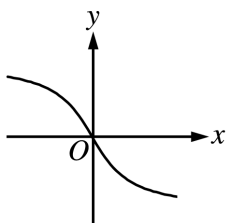
(A)



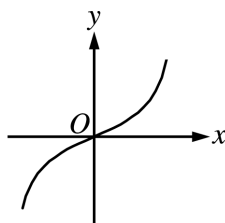
(B)



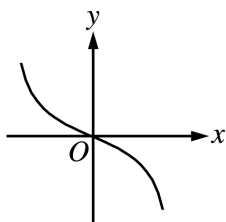
(C)



(D)



(E)



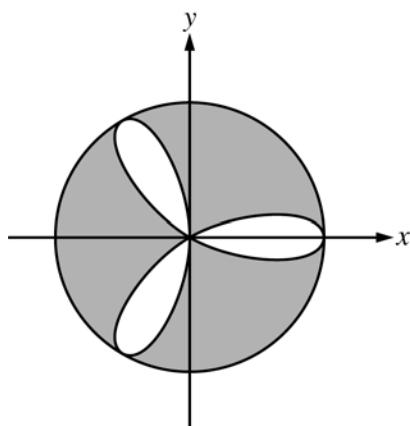
B**B****B****B****B****B****B****B****B**

89. A particle moves along a line so that its acceleration for $t \geq 0$ is given by $a(t) = \frac{t+3}{\sqrt{t^3+1}}$. If the particle's velocity at $t = 0$ is 5, what is the velocity of the particle at $t = 3$?

- (A) 0.713 (B) 1.134 (C) 6.134 (D) 6.710 (E) 11.710

90. If the series $\sum_{n=1}^{\infty} a_n$ converges and $a_n > 0$ for all n , which of the following must be true?

- (A) $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 0$
(B) $|a_n| < 1$ for all n
(C) $\sum_{n=1}^{\infty} a_n = 0$
(D) $\sum_{n=1}^{\infty} na_n$ diverges.
(E) $\sum_{n=1}^{\infty} \frac{a_n}{n}$ converges.

B**B****B****B****B****B****B****B****B**

91. The figure above shows the graphs of the polar curves $r = 2\cos(3\theta)$ and $r = 2$. What is the sum of the areas of the shaded regions?

(A) 0.858

(B) 3.142

(C) 8.566

(D) 9.425

(E) 15.708

B**B****B****B****B****B****B****B****B**

92. The function h is differentiable, and for all values of x , $h(x) = h(2 - x)$. Which of the following statements must be true?

I. $\int_0^2 h(x) \, dx > 0$

II. $h'(1) = 0$

III. $h'(0) = h'(2) = 1$

(A) I only

(B) II only

(C) III only

(D) II and III only

(E) I, II, and III

END OF SECTION I

**IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY
CHECK YOUR WORK ON PART B ONLY.**

DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
- **TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET**

**AFTER TIME HAS BEEN CALLED, TURN TO PAGE 38 AND
ANSWER QUESTIONS 93–96.**

CALCULUS BC
SECTION II, Part A
Time—30 minutes
Number of problems—2

A graphing calculator is required for these problems.

1

1

1

1

1

1

1

1

1

1

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

1. The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.
- (a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

-
- (b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

1**1****1****1****1****1****1****1****1****1**

- (c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

-
- (d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

2. For $t \geq 0$, a particle is moving along a curve so that its position at time t is $(x(t), y(t))$. At time $t = 2$, the particle is at position $(1, 5)$. It is known that $\frac{dx}{dt} = \frac{\sqrt{t+2}}{e^t}$ and $\frac{dy}{dt} = \sin^2 t$.

- (a) Is the horizontal movement of the particle to the left or to the right at time $t = 2$? Explain your answer. Find the slope of the path of the particle at time $t = 2$.

-
- (b) Find the x -coordinate of the particle's position at time $t = 4$.

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2**2****2****2****2****2****2****2****2****2**

(c) Find the speed of the particle at time $t = 4$. Find the acceleration vector of the particle at time $t = 4$.

(d) Find the distance traveled by the particle from time $t = 2$ to $t = 4$.

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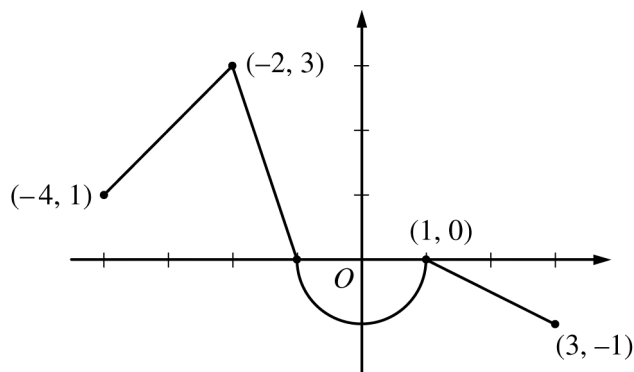
CALCULUS BC
SECTION II, Part B
Time—60 minutes
Number of problems—4

No calculator is allowed for these problems.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

GO ON TO THE NEXT PAGE.

NO CALCULATOR ALLOWED

Graph of f

3. Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

(a) Find the values of $g(2)$ and $g(-2)$.

(b) For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

3

3

3

3

3

3

3

3

3

3

NO CALCULATOR ALLOWED

- (c) Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.

-
- (d) For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

NO CALCULATOR ALLOWED

x	1	1.1	1.2	1.3	1.4
$f'(x)$	8	10	12	13	14.5

4. The function f is twice differentiable for $x > 0$ with $f(1) = 15$ and $f''(1) = 20$. Values of f' , the derivative of f , are given for selected values of x in the table above.

(a) Write an equation for the line tangent to the graph of f at $x = 1$. Use this line to approximate $f(1.4)$.

-
- (b) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_1^{1.4} f'(x) dx$. Use the approximation for $\int_1^{1.4} f'(x) dx$ to estimate the value of $f(1.4)$. Show the computations that lead to your answer.

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NO CALCULATOR ALLOWED

- (c) Use Euler's method, starting at $x = 1$ with two steps of equal size, to approximate $f(1.4)$. Show the computations that lead to your answer.

-
- (d) Write the second-degree Taylor polynomial for f about $x = 1$. Use the Taylor polynomial to approximate $f(1.4)$.

NO CALCULATOR ALLOWED

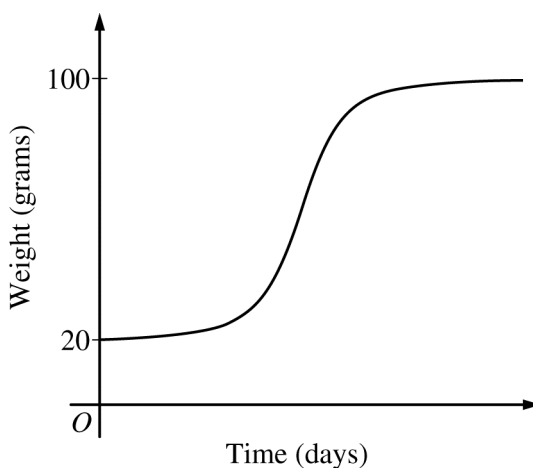
5. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

- (a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

- (b) Find $\frac{d^2B}{dt^2}$ in terms of B . Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



5**5****5****5****5****5****5****5****5****5****NO CALCULATOR ALLOWED**

- (c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

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NO CALCULATOR ALLOWED

6. The function g has derivatives of all orders, and the Maclaurin series for g is

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+3} = \frac{x}{3} - \frac{x^3}{5} + \frac{x^5}{7} - \cdots.$$

- (a) Using the ratio test, determine the interval of convergence of the Maclaurin series for g .

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NO CALCULATOR ALLOWED

- (b) The Maclaurin series for g evaluated at $x = \frac{1}{2}$ is an alternating series whose terms decrease in absolute value to 0. The approximation for $g\left(\frac{1}{2}\right)$ using the first two nonzero terms of this series is $\frac{17}{120}$. Show that this approximation differs from $g\left(\frac{1}{2}\right)$ by less than $\frac{1}{200}$.

-
- (c) Write the first three nonzero terms and the general term of the Maclaurin series for $g'(x)$.

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Answer Key for AP Calculus BC **Practice Exam, Section I**

Multiple-Choice Questions	
Question #	Key
1	E
2	A
3	B
4	A
5	C
6	C
7	A
8	C
9	D
10	E
11	A
12	C
13	C
14	E
15	A
16	C
17	C
18	A
19	C
20	C
21	E
22	D

23	A
24	E
25	B
26	B
27	C
28	D
76	B
77	B
78	C
79	E
80	E
81	D
82	C
83	C
84	D
85	B
86	B
87	B
88	E
89	E
90	E
91	D
92	B

AP[®] CALCULUS BC
2012 SCORING GUIDELINES

Question 1

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

- (a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.
- (b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.
- (c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.
- (d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

(a) $W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$
 $= 1.017$ (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

(b) $\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

(c) $\frac{1}{20} \int_0^{20} W(t) dt \approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15))$
 $= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9)$
 $= \frac{1}{20} \cdot 1215.8 = 60.79$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

(d) $W(25) = 71.0 + \int_{20}^{25} W'(t) dt$
 $= 71.0 + 2.043155 = 73.043$

2 : $\begin{cases} 1 : \text{estimate} \\ 1 : \text{interpretation with units} \end{cases}$

2 : $\begin{cases} 1 : \text{value} \\ 1 : \text{interpretation with units} \end{cases}$

3 : $\begin{cases} 1 : \text{left Riemann sum} \\ 1 : \text{approximation} \\ 1 : \text{underestimate with reason} \end{cases}$

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

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Question 2

For $t \geq 0$, a particle is moving along a curve so that its position at time t is $(x(t), y(t))$. At time $t = 2$, the particle is at position $(1, 5)$. It is known that $\frac{dx}{dt} = \frac{\sqrt{t+2}}{e^t}$ and $\frac{dy}{dt} = \sin^2 t$.

- (a) Is the horizontal movement of the particle to the left or to the right at time $t = 2$? Explain your answer. Find the slope of the path of the particle at time $t = 2$.
- (b) Find the x -coordinate of the particle's position at time $t = 4$.
- (c) Find the speed of the particle at time $t = 4$. Find the acceleration vector of the particle at time $t = 4$.
- (d) Find the distance traveled by the particle from time $t = 2$ to $t = 4$.

(a) $\left. \frac{dx}{dt} \right|_{t=2} = \frac{2}{e^2}$

Because $\left. \frac{dx}{dt} \right|_{t=2} > 0$, the particle is moving to the right at time $t = 2$.

$$\left. \frac{dy}{dx} \right|_{t=2} = \frac{\left. dy/dt \right|_{t=2}}{\left. dx/dt \right|_{t=2}} = 3.055 \text{ (or 3.054)}$$

3 : $\begin{cases} 1 : \text{moving to the right with reason} \\ 1 : \text{considers } \frac{dy/dt}{dx/dt} \\ 1 : \text{slope at } t = 2 \end{cases}$

(b) $x(4) = 1 + \int_2^4 \frac{\sqrt{t+2}}{e^t} dt = 1.253 \text{ (or 1.252)}$

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(c) Speed = $\sqrt{(x'(4))^2 + (y'(4))^2} = 0.575 \text{ (or 0.574)}$

$$\begin{aligned} \text{Acceleration} &= \langle x''(4), y''(4) \rangle \\ &= \langle -0.041, 0.989 \rangle \end{aligned}$$

2 : $\begin{cases} 1 : \text{speed} \\ 1 : \text{acceleration} \end{cases}$

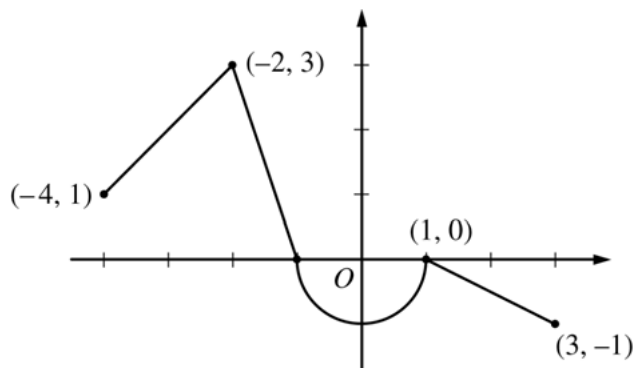
(d) Distance = $\int_2^4 \sqrt{(x'(t))^2 + (y'(t))^2} dt$
 $= 0.651 \text{ (or 0.650)}$

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

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Question 3

Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.



Graph of f

- (a) Find the values of $g(2)$ and $g(-2)$.
- (b) For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.
- (c) Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.
- (d) For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

(a) $g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right) = -\frac{1}{4}$

$$g(-2) = \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt$$

$$= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \frac{\pi}{2} - \frac{3}{2}$$

$$2 : \begin{cases} 1 : g(2) \\ 1 : g(-2) \end{cases}$$

(b) $g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$
 $g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$

$$2 : \begin{cases} 1 : g'(-3) \\ 1 : g''(-3) \end{cases}$$

- (c) The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$.
 Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

$$3 : \begin{cases} 1 : \text{considers } g'(x) = 0 \\ 1 : x = -1 \text{ and } x = 1 \\ 1 : \text{answers with justifications} \end{cases}$$

- (d) The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

$$2 : \begin{cases} 1 : \text{answer} \\ 1 : \text{explanation} \end{cases}$$

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Question 4

x	1	1.1	1.2	1.3	1.4
$f'(x)$	8	10	12	13	14.5

The function f is twice differentiable for $x > 0$ with $f(1) = 15$ and $f''(1) = 20$. Values of f' , the derivative of f , are given for selected values of x in the table above.

- (a) Write an equation for the line tangent to the graph of f at $x = 1$. Use this line to approximate $f(1.4)$.
- (b) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate $\int_1^{1.4} f'(x) dx$. Use the approximation for $\int_1^{1.4} f'(x) dx$ to estimate the value of $f(1.4)$. Show the computations that lead to your answer.
- (c) Use Euler's method, starting at $x = 1$ with two steps of equal size, to approximate $f(1.4)$. Show the computations that lead to your answer.
- (d) Write the second-degree Taylor polynomial for f about $x = 1$. Use the Taylor polynomial to approximate $f(1.4)$.

(a) $f(1) = 15$, $f'(1) = 8$

An equation for the tangent line is
 $y = 15 + 8(x - 1)$.

$$f(1.4) \approx 15 + 8(1.4 - 1) = 18.2$$

$$2 : \begin{cases} 1 : \text{tangent line} \\ 1 : \text{approximation} \end{cases}$$

(b) $\int_1^{1.4} f'(x) dx \approx (0.2)(10) + (0.2)(13) = 4.6$

$$f(1.4) = f(1) + \int_1^{1.4} f'(x) dx$$

$$f(1.4) \approx 15 + 4.6 = 19.6$$

$$3 : \begin{cases} 1 : \text{midpoint Riemann sum} \\ 1 : \text{Fundamental Theorem of Calculus} \\ 1 : \text{answer} \end{cases}$$

(c) $f(1.2) \approx f(1) + (0.2)(8) = 16.6$

$$f(1.4) \approx 16.6 + (0.2)(12) = 19.0$$

$$2 : \begin{cases} 1 : \text{Euler's method with two steps} \\ 1 : \text{answer} \end{cases}$$

(d) $T_2(x) = 15 + 8(x - 1) + \frac{20}{2!}(x - 1)^2$

$$= 15 + 8(x - 1) + 10(x - 1)^2$$

$$f(1.4) \approx 15 + 8(1.4 - 1) + 10(1.4 - 1)^2 = 19.8$$

$$2 : \begin{cases} 1 : \text{Taylor polynomial} \\ 1 : \text{approximation} \end{cases}$$

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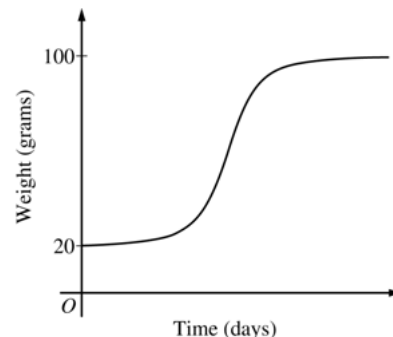
Question 5

The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

- (a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.
- (b) Find $\frac{d^2B}{dt^2}$ in terms of B . Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.
- (c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.



(a) $\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

(b) $\frac{d^2B}{dt^2} = -\frac{1}{5} \frac{dB}{dt} = -\frac{1}{5} \cdot \frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

(c) $\frac{dB}{dt} = \frac{1}{5}(100 - B)$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \frac{1}{5}t + C$$

Because $20 \leq B < 100$, $|100 - B| = 100 - B$.

$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5}, \quad t \geq 0$$

$$2 : \begin{cases} 1 : \text{uses } \frac{dB}{dt} \\ 1 : \text{answer with reason} \end{cases}$$

$$2 : \begin{cases} 1 : \frac{d^2B}{dt^2} \text{ in terms of } B \\ 1 : \text{explanation} \end{cases}$$

$$5 : \begin{cases} 1 : \text{separation of variables} \\ 1 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ 1 : \text{uses initial condition} \\ 1 : \text{solves for } B \end{cases}$$

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

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Question 6

The function g has derivatives of all orders, and the Maclaurin series for g is

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+3} = \frac{x}{3} - \frac{x^3}{5} + \frac{x^5}{7} - \cdots$$

- (a) Using the ratio test, determine the interval of convergence of the Maclaurin series for g .
- (b) The Maclaurin series for g evaluated at $x = \frac{1}{2}$ is an alternating series whose terms decrease in absolute value to 0. The approximation for $g\left(\frac{1}{2}\right)$ using the first two nonzero terms of this series is $\frac{17}{120}$. Show that this approximation differs from $g\left(\frac{1}{2}\right)$ by less than $\frac{1}{200}$.
- (c) Write the first three nonzero terms and the general term of the Maclaurin series for $g'(x)$.

(a) $\left| \frac{x^{2n+3}}{2n+5} \cdot \frac{2n+3}{x^{2n+1}} \right| = \left(\frac{2n+3}{2n+5} \right) \cdot x^2$

$$\lim_{n \rightarrow \infty} \left(\frac{2n+3}{2n+5} \right) \cdot x^2 = x^2$$

$$x^2 < 1 \Rightarrow -1 < x < 1$$

The series converges when $-1 < x < 1$.

When $x = -1$, the series is $-\frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots$

This series converges by the Alternating Series Test.

When $x = 1$, the series is $\frac{1}{3} - \frac{1}{5} + \frac{1}{7} - \frac{1}{9} + \cdots$

This series converges by the Alternating Series Test.

Therefore, the interval of convergence is $-1 \leq x \leq 1$.

5 : $\begin{cases} 1 : \text{sets up ratio} \\ 1 : \text{computes limit of ratio} \\ 1 : \text{identifies interior of} \\ \quad \text{interval of convergence} \\ 1 : \text{considers both endpoints} \\ 1 : \text{analysis and interval of convergence} \end{cases}$

(b) $\left| g\left(\frac{1}{2}\right) - \frac{17}{120} \right| < \frac{\left(\frac{1}{2}\right)^5}{7} = \frac{1}{224} < \frac{1}{200}$

2 : $\begin{cases} 1 : \text{uses the third term as an error bound} \\ 1 : \text{error bound} \end{cases}$

(c) $g'(x) = \frac{1}{3} - \frac{3}{5}x^2 + \frac{5}{7}x^4 + \cdots + (-1)^n \left(\frac{2n+1}{2n+3} \right) x^{2n} + \cdots$

2 : $\begin{cases} 1 : \text{first three terms} \\ 1 : \text{general term} \end{cases}$